

Additive Quotient Tomography of the Identity Bucket: Minimal Fourier Recovery, Folded Parseval, and the Twist–Stable Möbius Gate

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Abstract

TPC-38 showed that a three-modulus multiplicative completion identifies the recombined output with the joint–alias sum, not its original equality column. Multiplicative characters cannot distinguish those aliases because every one lies in the same ratio bucket. We construct the missing coefficient–free recovery map on the additive quotient of that bucket.

Let M be the three-modulus CRT product, let $|F| \leq B$, and put

$$L = \left\lfloor \frac{B}{M} \right\rfloor, \quad q = L + 1, \quad \omega = \exp(2\pi i / q).$$

Then $qM > B$, and on the full determinant window

$$\mathbf{1}_{M|F} \frac{1}{q} \sum_{\nu=0}^{q-1} \exp(2\pi i \nu F / (qM)) = \mathbf{1}_{qM|F} = \mathbf{1}_{F=0}.$$

Full/proper face recombination is essential in this identity: the additive average alone does not remove determinants with $M \nmid F$. For arbitrary Hilbert columns $\mathcal{Z}_k$, $-L \leq k \leq L$, define

$$\mathcal{Y}_\nu = \sum_{k=-L}^L \omega^{\nu k} \mathcal{Z}_k.$$

The quotient projector gives the exact recovery formula with unit normalized amplification

$$\mathcal{Z}_0 = \frac{1}{q} \sum_{\nu=0}^{q-1} \mathcal{Y}_\nu, \quad \|\mathcal{Z}_0\|^2 \leq \frac{1}{q} \sum_{\nu=0}^{q-1} \|\mathcal{Y}_\nu\|^2.$$

We prove the Hilbert–valued folded Parseval identity

$$\frac{1}{q} \sum_{\nu=0}^{q-1} \|\mathcal{Y}_\nu\|^2 = \|\mathcal{Z}_0\|^2 + \sum_{r=1}^L \|\mathcal{Z}_r + \mathcal{Z}_{r-q}\|^2.$$

The normalized scalar alias–index tomography Gram has eigenvalues 2, 1, 0, with multiplicities $L, 1, L$, respectively. Its kernel consists precisely of antisymmetric folded alias pairs and does not intersect the equality column. Thus the minimal bank is weaker than a raw alias trace or a full Gram estimate while still recovering the one column required by the physical problem.

The number $q = L + 1$ is optimal. Even allowing arbitrary nonzero phase nodes and complex recovery weights, no bank of at most L phase measurements recovers the central coefficient of every Laurent polynomial supported on $[-L, L]$. The regular q -gon attains this lower bound and also minimizes the normalized noise amplification. Under positive weights,

every minimally sampled optimal rule is a rotation of this regular polygon with equal weights. Deleting any one twist from the minimal bank destroys coefficient-free exact recovery.

For the orbit-sliced prime-Möbius interface, the twist is $\exp(2\pi i\nu(su - N)/(qM))$. At the endpoint,

$$q \asymp 1 + X/M = X^{1/400+o(1)}, \quad Q^2 J q = X^{o(1)} Q^3 / J.$$

Consequently, a square-average bound of this target size for the recombined twisted outputs would close the inherited equality gate with no additional alias loss. The corresponding full/proper fields have exactly the same coefficient masses as their untwisted versions. However, folded Parseval retains the original equality energy as a positive summand, and the terminal expansion retains affine four-Möbius correlations.

We therefore prove an exact, minimally sampled equality projector and a new twist-stable conditional route, not the required Möbius energy estimate. No affine Chowla theorem, parity breach, Hardy-Littlewood asymptotic, or twin-prime theorem is claimed.

Keywords: Möbius correlation; additive character; quotient tomography; CRT completion; folded Parseval; alias recovery; parity boundary.

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1 Introduction

The orbit-sliced program in TPC-34-TPC-38 reduces a prime-pair remainder to a coefficient-specific Hilbert energy for reflected Möbius divisor sums [4–8]. After exact full/proper recombination, three orbit-scale CRT coordinates cancel all nonjoint determinant ghosts. The remaining congruence completion is

$$\mathcal{Y}_{\text{alias}} = \sum_{k=-L}^L \mathcal{Z}_k, \quad L = \left\lfloor \frac{B}{M} \right\rfloor, \quad (1)$$

where M is the three-modulus product, $|F| \leq B$ is the determinant window, and $\mathcal{Z}_k$ is the raw column on $F = kM$. The physical equality output is only

$$\mathcal{Y}_{\text{phys}} = \mathcal{Z}_0. \quad (2)$$

TPC-38 proved that (1) does not control (2) in the unrestricted Hilbert column class. The obstruction is elementary and sharp: the columns $\mathcal{Z}_0 = v$ and $\mathcal{Z}_1 = -v$ make the completed sum vanish while leaving the equality column arbitrary. Multiplicative character refinement cannot separate them, since every joint alias satisfies

$$(N + kM)N^{-1} \equiv 1 \pmod{M}. \quad (3)$$

The present paper uses the additive quotient that remains after (3). Put

$$q = L + 1, \quad \omega = \exp(2\pi i 1/q). \quad (4)$$

Then the vectors

$$\mathcal{Y}_\nu = \sum_{k=-L}^L \omega^{\nu k} \mathcal{Z}_k, \quad 0 \leq \nu < q, \quad (5)$$

obey

$$\boxed{\mathcal{Z}_0 = \frac{1}{q} \sum_{\nu=0}^{q-1} \mathcal{Y}_\nu.} \quad (6)$$

This is not a full discrete Fourier inversion: there are only $q = L + 1$ measurements for $2L + 1$ alias columns. Positive and negative aliases are folded together modulo q , but the zero residue class contains only $k = 0$. This is exactly the amount of information needed for equality recovery.

1.1 The finite quotient projector

The physical form of (6) is the identity

$$\mathbf{1}_{M|F} \frac{1}{q} \sum_{\nu=0}^{q-1} \exp(2\pi i \nu F / (qM)) = \mathbf{1}_{F=0} \quad (|F| \leq B). \quad (7)$$

Indeed, the left side is $\mathbf{1}_{qM|F}$, and $qM > B$. The factor $\mathbf{1}_{M|F}$ is supplied by the exact full/proper recombination

$$C_N(F) + P_N(F) = \mathbf{1}_{M|F} \quad (8)$$

proved in TPC–38. It cannot be omitted: the additive average alone is generally nonzero when $M \nmid F$. Only after full/proper recombination restricts the determinant to $F = kM$ does the quotient average select the multiples of qM .

Thus (7) is an additive projector on the quotient

$$M\mathbb{Z}/qM\mathbb{Z} \cong \mathbb{Z}/q\mathbb{Z}. \quad (9)$$

It converts the congruence completion into exact equality without adding a fourth multiplicative character coordinate.

1.2 Folded Parseval and the exact spectrum

For Hilbert-valued columns, character orthogonality on (9) gives

$$\frac{1}{q} \sum_{\nu=0}^{q-1} \|\mathcal{Y}_\nu\|^2 = \|\mathcal{Z}_0\|^2 + \sum_{r=1}^L \|\mathcal{Z}_r + \mathcal{Z}_{r-q}\|^2. \quad (10)$$

The associated normalized tomography Gram has spectrum

$$2 \quad (L \text{ times}), \quad 1 \quad (1 \text{ time}), \quad 0 \quad (L \text{ times}). \quad (11)$$

The eigenvalue-one direction is the zero alias. The kernel consists of antisymmetric pairs

$$\mathcal{Z}_r = -\mathcal{Z}_{r-q}, \quad 1 \leq r \leq L, \quad (12)$$

which do not affect $\mathcal{Z}_0$. Hence this bank is strictly weaker than a full alias trace and much weaker than an alias-Gram operator bound, but it is sufficient for the required column.

1.3 Minimal sampling and optimal recovery amplification

The sample count in (4) is best possible. Suppose phase nodes $z_1, \dots, z_m \in \mathbb{C}^\times$ and complex recovery weights are required to recover the central coefficient of every Laurent polynomial supported on $[-L, L]$. If $m \leq L$, the polynomial

$$Q(z) = \prod_{j=1}^m (z - z_j) \quad (13)$$

has degree at most L , vanishes at every measurement node, and has nonzero constant coefficient. Exact recovery is therefore impossible. The regular $(L+1)$ -gon attains the lower bound.

If a is the recovery vector for m measurements, its normalized noise amplification is

$$\mathcal{A} = m \|a\|_2^2. \quad (14)$$

The constant moment forces $\mathcal{A} \geq 1$. Equal weights on the regular polygon give $\mathcal{A} = 1$. Under positive weights, the minimally sampled rule is rigid: the weights are equal and the nodes are a rotation of the regular polygon.

1.4 The new physical gate

For the physical determinant $F = su - N$, define the twisted face fields

$$C_{\nu,N}(F) = \exp(2\pi i \nu F / (qM)) C_N(F), \quad P_{\nu,N}(F) = \exp(2\pi i \nu F / (qM)) P_N(F). \quad (15)$$

Their coefficient masses are exactly those of the untwisted fields. After recombination, their outputs are the vectors in (5). Therefore the explicit conditional input

$$\frac{1}{q} \sum_{\nu=0}^{q-1} \|\mathcal{Y}_\nu\|^2 \ll X^{o(1)} Q^2 J(2L+1) \quad (16)$$

implies the physical equality gate. At the endpoint,

$$2L+1 \asymp q \asymp X^{1/400+o(1)}, \quad Q^2 J(2L+1) = X^{o(1)} Q^3 / J. \quad (17)$$

Equation (16) is not proved here. Folded Parseval shows that it retains $\|\mathcal{Z}_0\|^2$ as a positive summand. In the terminal factorization $s = 1$, the twists create a short Fourier transform along the progression $N + kM$, but the square still contains the original affine four-Möbius term and long-gap correlations. Additive tomography repairs the logical recovery gap; it does not manufacture arithmetic cancellation.

1.5 Organization and claim boundary

Section 2 states the inherited physical interface. Section 3 proves the quotient projector and exact Hilbert recovery. Section 4 proves folded Parseval and the complete spectrum. Section 5 proves sample minimality, recovery-amplification optimality, and positive-rule rigidity. Section 6 compares phase recovery with pure divisibility completions. Section 7 formulates the twist-stable physical gate. Section 8 records its terminal Möbius content. Section 9 treats partial and random banks, and Section 10 gives the corrected route tree.

All quotient, Fourier, spectrum, minimality, and finite divisibility results are unconditional. The paper proves no twist-stable physical energy estimate, no new Möbius autocorrelation estimate, no affine Chowla theorem, no failure of sieve parity, no Hardy-Littlewood prime-pair asymptotic, and no twin-prime theorem.

2 Inherited physical interface and alias notation

We use the physical packet and exact CRT face calculus of TPC-37 and TPC-38. This section restates only the data needed to distinguish the unconditional tomography from its conditional arithmetic application.

2.1 Rows, targets, and the equality output

Fix $h \in \mathbb{Z} \setminus \{0\}$. On a retained dyadic packet,

$$Q = L_0 D_0, \quad QJ \asymp X, \quad (18)$$

where L_0 and D_0 denote the two source scales, so that L remains reserved for the alias radius. A physical row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad \ell_\alpha \asymp L_0, \quad d_\alpha \asymp D_0, \quad m_\alpha = \ell_\alpha d_\alpha \asymp Q, \quad (19)$$

and the source separation makes m_α determine α . The two row coefficient systems obey

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}|^2 \ll X^{o(1)} Q, \quad i = 1, 2. \quad (20)$$

For j in a positive interval $\mathcal{J}$ of $O(J)$ integers, put

$$N_\alpha(j) = m_\alpha j + h \asymp X. \quad (21)$$

The retained output space is

$$\mathcal{H}_{\text{phys}} = \ell^2\{(L, \gamma, j)\} \oplus \ell^2\{(R, \alpha, j)\}. \quad (22)$$

For a determinant shift F , let $\mathcal{Z}_F[\kappa] \in \mathcal{H}_{\text{phys}}$ denote the shifted output on

$$su = N_\alpha(j) + F, \quad (23)$$

with target atom multiplier $\kappa_N(F)$. Write

$$\mathcal{R}_F := \mathcal{Z}_F[\mathbf{1}] \quad (24)$$

for the raw column. The original regular-subpacket output is

$$\mathcal{Y}_{\text{phys}} = \mathcal{R}_0, \quad \|\mathcal{R}_0\|^2 = V_L + V_R. \quad (25)$$

The inherited target is

$$\boxed{\|\mathcal{R}_0\|^2 \ll X^{o(1)} \frac{Q^3}{J}}. \quad (26)$$

2.2 The three-modulus completion

Choose distinct primes $q_1, q_2, q_3 \asymp J$ satisfying the inherited range conditions

$$\max \mathcal{J} < q_i, \quad \max_\alpha d_\alpha < q_i < \min_\alpha \ell_\alpha, \quad q_i \nmid h, \quad (27)$$

and put

$$M = q_1 q_2 q_3. \quad (28)$$

TPC-37 absorbs the row-time pairs for which $(N_\alpha(j), M) > 1$ at the inherited acceptable scale. All physical outputs below refer to the complementary regular subpacket. No claim is made that the additive twists introduced later supply this zero-face estimate; it is inherited before completion.

TPC-38 constructs the full and proper target fields $C_N(F)$ and $P_N(F)$. Their exact recombination is

$$\boxed{C_N(F) + P_N(F) = \mathbf{1}_{M|F}}. \quad (29)$$

Let B be an integer such that every determinant used by the completed packet satisfies $|F| \leq B$. We assume $B \geq M$, equivalently $L \geq 1$, as is automatic in the endpoint regime considered below, and put

$$L = \left\lfloor \frac{B}{M} \right\rfloor, \quad \mathcal{I}_B = \{-L, \dots, L\}, \quad K_B = 2L + 1. \quad (30)$$

For $k \in \mathcal{I}_B$, define

$$\mathcal{Z}_k := \mathcal{R}_{kM}. \quad (31)$$

Summing (29) before estimating gives

$$\sum_{|F| \leq B} (\mathcal{Z}_F[C] + \mathcal{Z}_F[P]) = \sum_{k=-L}^L \mathcal{Z}_k. \quad (32)$$

The right side is a congruence completion, not $\mathcal{Z}_0$.

2.3 Endpoint normalization

At the endpoint,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad M = X^{399/400+o(1)}. \quad (33)$$

For $B \asymp X$,

$$K_B \asymp 1 + \frac{X}{M} = X^{1/400+o(1)}. \quad (34)$$

Put

$$B_0 = Q^2 J. \quad (35)$$

Then

$$\boxed{B_0 K_B = X^{o(1)} \frac{Q^3}{J}.} \quad (36)$$

The proved input-copy diagonal has size $X^{o(1)} B_0$, but it is not a bound for the raw output column $\mathcal{Z}_0$. The latter remains the physical gate in (26).

3 The exact additive quotient projector

The multiplicative CRT coordinates have already done their job once (29) is reached. The surviving problem is finite additive recovery on the alias index.

3.1 The minimal no-wrap quotient

Retain the notation in (30) and put

$$q = L + 1, \quad \omega = \exp(2\pi i / q). \quad (37)$$

The integer q need not be prime.

Lemma 3.1 (Minimal quotient size). *One has*

$$B < qM \leq B + M. \quad (38)$$

Moreover, the residue class $0 \pmod{q}$ meets $\{-L, \dots, L\}$ only at 0.

Proof. Write $B = LM + r$, where $0 \leq r < M$. Then $(L + 1)M = B + (M - r)$, proving (38). Since $q = L + 1$, every nonzero k in the alias interval satisfies $0 < |k| < q$. $\square$

Theorem 3.2 (Additive quotient identity). *For every integer F with $|F| \leq B$,*

$$\boxed{\mathbf{1}_{M|F} \frac{1}{q} \sum_{\nu=0}^{q-1} \exp(2\pi i \nu F / (qM)) = \mathbf{1}_{qM|F} = \mathbf{1}_{F=0}.} \quad (39)$$

Proof. If $M \nmid F$, the first member is zero. If $F = kM$, additive orthogonality gives

$$\frac{1}{q} \sum_{\nu=0}^{q-1} \omega^{\nu k} = \mathbf{1}_{q|k}.$$

Thus the first member is $\mathbf{1}_{qM|F}$. By lemma 3.1, the only such determinant in the window is $F = 0$. $\square$

The first factor in (39) is essential in the physical construction. The additive average by itself is generally nonzero when $M \nmid F$, so it is not a divisibility indicator on the full determinant window. The product with $\mathbf{1}_{M|F}$ is the indicator $\mathbf{1}_{qM|F}$. The available completed objects are the separate full and proper face outputs; their nonjoint terms disappear only after using (29) with the same twist on both fields.

3.2 Twisted face outputs

For $0 \leq \nu < q$, define

$$\mathcal{Y}_\nu^C := \sum_{|F| \leq B} \mathcal{Z}_F[\exp(2\pi i \nu F / (qM)) C], \quad (40)$$

$$\mathcal{Y}_\nu^P := \sum_{|F| \leq B} \mathcal{Z}_F[\exp(2\pi i \nu F / (qM)) P], \quad (41)$$

$$\mathcal{Y}_\nu := \mathcal{Y}_\nu^C + \mathcal{Y}_\nu^P. \quad (42)$$

Proposition 3.3 (Recombined twist formula). *For every ν ,*

$$\boxed{\mathcal{Y}_\nu = \sum_{k=-L}^L \omega^{\nu k} \mathcal{Z}_k.} \quad (43)$$

In particular, all nonjoint face atoms cancel before the alias bank is formed.

Proof. Linearity and (29) give

$$\mathcal{Y}_\nu = \sum_{|F| \leq B} \exp(2\pi i \nu F / (qM)) \mathbf{1}_{M|F} \mathcal{R}_F.$$

Write $F = kM$ and use (31). □

Theorem 3.4 (Exact Hilbert equality recovery). *For arbitrary columns $\mathcal{Z}_k \in \mathcal{H}_{\text{phys}}$,*

$$\boxed{\mathcal{Z}_0 = \frac{1}{q} \sum_{\nu=0}^{q-1} \mathcal{Y}_\nu.} \quad (44)$$

Consequently,

$$\boxed{\|\mathcal{Z}_0\|^2 \leq \frac{1}{q} \sum_{\nu=0}^{q-1} \|\mathcal{Y}_\nu\|^2.} \quad (45)$$

Proof. Average (43) over ν . By lemma 3.1, only $k = 0$ survives. Jensen's inequality gives (45). □

There is no factor q in (45). The recovery functional is the orthogonal average of a regular polygon bank. The later minimality theorem shows that this sample count and normalized recovery amplification are simultaneously optimal. This is not a condition-number assertion for the full tomography map, which has a nontrivial kernel.

Remark 3.5 (What has and has not been recovered). The equality column is recovered exactly from the *values* of all twisted completed outputs. No bound for those outputs has yet been proved. Equation (45) converts a twist-stable energy theorem into the desired physical estimate; it is not itself such an energy theorem.

4 Hilbert-valued folded Parseval and tomography spectrum

The minimal quotient deliberately identifies some positive and negative aliases. This section computes that folding exactly.

4.1 General quotient sizes

Let

$$L < r \leq 2L + 1 \quad (46)$$

and put $\zeta_r = \exp(2\pi i/r)$. For Hilbert columns $Z_{-L}, \dots, Z_L \in \mathcal{H}$, define

$$Y_\nu^{(r)} := \sum_{k=-L}^L \zeta_r^{\nu k} Z_k, \quad 0 \leq \nu < r, \quad (47)$$

$$W_a^{(r)} := \sum_{\substack{-L \leq k \leq L \\ k \equiv a \pmod{r}}} Z_k, \quad a \in \mathbb{Z}/r\mathbb{Z}. \quad (48)$$

Theorem 4.1 (Hilbert-valued folded Parseval). *For every r in (46),*

$$\frac{1}{r} \sum_{\nu=0}^{r-1} \|Y_\nu^{(r)}\|^2 = \sum_{a \bmod r} \|W_a^{(r)}\|^2. \quad (49)$$

At the minimal value $r = q = L + 1$,

$$\frac{1}{q} \sum_{\nu=0}^{q-1} \|Y_\nu^{(q)}\|^2 = \|Z_0\|^2 + \sum_{a=1}^L \|Z_a + Z_{a-q}\|^2. \quad (50)$$

Proof. Expand the squared norms and average the phase $\zeta_r^{\nu(k-k')}$. Additive orthogonality retains precisely the pairs $k \equiv k' \pmod{r}$, proving (49). If $r = L + 1$, the zero residue class is $\{0\}$, while residue a , $1 \leq a \leq L$, contains the pair $\{a, a - q\}$. $\square$

4.2 Complete singular spectrum

Define the normalized tomography operator

$$\mathcal{T}_r : \mathcal{H}^{2L+1} \longrightarrow \mathcal{H}^r, \quad (\mathcal{T}_r Z)_\nu = r^{-1/2} \sum_{k=-L}^L \zeta_r^{\nu k} Z_k. \quad (51)$$

Its scalar Gram matrix is

$$(\mathcal{T}_r^* \mathcal{T}_r)_{k,k'} = \mathbf{1}_{k \equiv k' \pmod{r}}. \quad (52)$$

Theorem 4.2 (Alias-index tomography spectrum). *For $L < r \leq 2L + 1$, the spectrum of the scalar alias-index Gram matrix in (52) is*

eigenvalue	multiplicity
2	$2L + 1 - r$
1	$2r - (2L + 1)$
0	$2L + 1 - r$.

(53)

At $r = q = L + 1$, this becomes

$$2 \text{ (} L \text{ times),} \quad 1 \text{ (1 time),} \quad 0 \text{ (} L \text{ times).} \quad (54)$$

At the minimal value $r = q$, the eigenvalue-one subspace is the isolated zero column. The kernel is

$$Z_0 = 0, \quad Z_a = -Z_{a-q}, \quad 1 \leq a \leq L. \quad (55)$$

As an operator on $\mathcal{H}^{2L+1}$, one has $\mathcal{T}_r^* \mathcal{T}_r = G_r \otimes I_{\mathcal{H}}$, where G_r is this scalar matrix. Thus for finite $\dim \mathcal{H} = d$, every displayed multiplicity is multiplied by d ; the table records the intrinsic alias-index multiplicities.

Proof. Because $r > L$, every residue bucket in (48) has size one or two. There are $2L + 1 - r$ doubletons and $2r - (2L + 1)$ singletons. On a bucket of size s , the Gram block in (52) is the all-ones matrix J_s , whose eigenvalues are s once and zero $s - 1$ times. This proves (53). The minimal specialization follows from the explicit buckets in the proof of theorem 4.1. $\square$

At the other endpoint $r = 2L + 1$, every bucket is a singleton and

$$\frac{1}{2L + 1} \sum_{\nu=0}^{2L} \|Y_\nu^{(2L+1)}\|^2 = \sum_{k=-L}^L \|Z_k\|^2. \quad (56)$$

Thus $L + 1$ twists are sufficient to recover the zero column, whereas $2L + 1$ twists are required to separate every alias column by this cyclic construction.

4.3 Exact position between trace and Gram gates

Put

$$E_{\text{tr}}(Z) := \sum_{k=-L}^L \|Z_k\|^2, \quad (57)$$

$$E_{\text{fold}}(Z) := \|Z_0\|^2 + \sum_{a=1}^L \|Z_a + Z_{a-q}\|^2. \quad (58)$$

Then

$$\|Z_0\|^2 \leq E_{\text{fold}}(Z) \leq 2E_{\text{tr}}(Z). \quad (59)$$

If $\mathbf{G} = (\langle Z_k, Z_{k'} \rangle)$, then

$$E_{\text{tr}}(Z) = \text{tr } \mathbf{G} \leq (2L + 1) \|\mathbf{G}\|_{\text{op}}. \quad (60)$$

Consequently,

$$\text{alias Gram} \implies \text{raw trace} \implies \text{minimal folded energy} \implies \text{zero-column energy}. \quad (61)$$

The reverse implications fail in the unrestricted Hilbert class. Taking $Z_a = Tv$, $Z_{a-q} = -Tv$, and all other columns zero makes $E_{\text{fold}} = 0$ while the trace and Gram norm grow like T^2 . Taking the same pair with equal signs and $Z_0 = 0$ makes the zero-column energy vanish while E_{fold} is arbitrarily large. Hence the minimal tomography bank discards exactly a large collection of directions that are irrelevant to equality recovery; it is not a disguised full trace theorem.

5 Minimal sampling, optimal recovery amplification, and rigidity

The root-of-unity construction uses only half as many measurements as a full Fourier inversion. We now show that no phase bank can use fewer.

5.1 Universal phase measurements

Let $z \in \mathbb{C}^\times$. For a Hilbert column family $Z = (Z_{-L}, \dots, Z_L)$, define the phase measurement

$$\mathcal{Y}_Z(z) = \sum_{k=-L}^L z^k Z_k. \quad (62)$$

Given nodes $z_1, \dots, z_m$ and coefficients $a_1, \dots, a_m$, the rule

$$Z_0 = \sum_{j=1}^m a_j \mathcal{Y}_Z(z_j) \quad (63)$$

holds for every Hilbert space and every column family if and only if

$$\sum_{j=1}^m a_j z_j^k = \mathbf{1}_{k=0} \quad (-L \leq k \leq L). \quad (64)$$

Theorem 5.1 (Minimal phase count). *If (63) holds universally, then*

$$\boxed{m \geq L + 1}. \quad (65)$$

This remains true with arbitrary nonzero complex nodes and arbitrary complex recovery weights.

Proof. Suppose $m \leq L$ and form

$$Q(z) = \prod_{j=1}^m (z - z_j) = \sum_{k=0}^m c_k z^k. \quad (66)$$

Its constant coefficient is $c_0 = (-1)^m \prod_j z_j \neq 0$. Take scalar columns $Z_k = c_k$ for $0 \leq k \leq m$ and zero otherwise. Then every measurement in (63) is $Q(z_j) = 0$, whereas $Z_0 = c_0 \neq 0$, a contradiction. $\square$

Corollary 5.2 (Optimality of the quotient bank). *The nodes*

$$z_\nu = \omega^\nu, \quad 0 \leq \nu < q = L + 1, \quad (67)$$

with weights $a_\nu = 1/q$ attain the lower bound in theorem 5.1.

Proof. The moment system (64) is additive orthogonality modulo q , and $q > |k|$ for every nonzero alias index. $\square$

The lower bound concerns recovery of only Z_0 . Recovering every one of the $2L + 1$ arbitrary columns requires a measurement matrix of rank $2L + 1$, and therefore at least $2L + 1$ samples. The full DFT in (56) attains that different lower bound.

5.2 Noise amplification

Assume now that $z_j \in \mathbb{T}$. For a recovery rule satisfying (64), define

$$\mathcal{A}(a) := m \|a\|_2^2. \quad (68)$$

The Cauchy bound associated with the rule is

$$\|Z_0\|^2 \leq \|a\|_2^2 \sum_{j=1}^m \|\mathcal{Y}_Z(z_j)\|^2. \quad (69)$$

Proposition 5.3 (Optimal normalized amplification). *Every universal recovery rule satisfies*

$$\boxed{\mathcal{A}(a) \geq 1}. \quad (70)$$

The regular polygon rule in corollary 5.2 has $\mathcal{A} = 1$.

Proof. The moment at $k = 0$ says $\sum_j a_j = 1$. Hence $1 \leq m \sum_j |a_j|^2$. Equal weights $1/q$ give equality. $\square$

Thus the quotient bank is simultaneously sample-minimal and optimal for the normalized recovery amplification. Its estimate (45) is the equality case of (69) after the natural average normalization. The full tomography operator is singular, so this statement must not be read as a condition number for that operator.

5.3 Rigidity of positive minimal rules

The regular polygon is not merely one positive optimal construction.

Theorem 5.4 (Positive-rule rigidity). *Let $m = q = L+1$, let $z_1, \dots, z_q \in \mathbb{T}$, and let $w_1, \dots, w_q > 0$. Suppose*

$$\sum_{j=1}^q w_j z_j^k = \mathbf{1}_{k=0} \quad (-L \leq k \leq L). \quad (71)$$

Then

$$w_j = \frac{1}{q} \quad (1 \leq j \leq q), \quad (72)$$

and, after reordering, there is $\rho \in \mathbb{T}$ such that

$$\{z_1, \dots, z_q\} = \{\rho \omega^\nu : 0 \leq \nu < q\}. \quad (73)$$

Proof. Form the square matrix

$$U_{j,r} = \sqrt{w_j} z_j^r, \quad 1 \leq j \leq q, \quad 0 \leq r \leq q-1. \quad (74)$$

By (71), $(U^*U)_{r,s} = \delta_{r,s}$, since $|s-r| \leq q-1 = L$. Thus U is unitary. The squared norm of row j is qw_j , which proves (72). Orthogonality of two distinct rows gives

$$\sum_{r=0}^{q-1} (z_j \bar{z}_i)^r = 0.$$

Therefore every quotient $z_j \bar{z}_i$ is a nontrivial q -th root of unity. The q distinct nodes are one rotation of all q -th roots. $\square$

5.4 Every minimal twist is necessary

Corollary 5.5 (Deletion no-go). *Delete any one measurement from the minimal regular polygon bank. There are scalar alias columns with*

$$\mathcal{Y}_Z(z_\nu) = 0 \quad \text{at every retained node,} \quad Z_0 \neq 0. \quad (75)$$

Hence estimates for all but one minimal twist do not give coefficient-free exact equality recovery.

Proof. There remain L nodes. Use the annihilating polynomial in (66). $\square$

If an analytic argument has exceptional frequencies, the recovery bank must therefore be made redundant or the exceptional outputs must be controlled separately. Treating an unproved minimal twist as negligible is not a valid recovery step.

6 Divisibility completions and the alias-synthesis trade

The quotient projector can be compared with the multi-completion route proposed in TPC-38. The comparison explains both why a superwindow modulus is unavoidable and why additive phases are a distinct implementation of it.

6.1 Exact rigidity of linear divisibility recovery

For a positive integer m , put

$$d_m(F) = \mathbf{1}_{m|F}, \quad |F| \leq B. \quad (76)$$

Repeated moduli in a linear combination are combined into one aggregate coefficient a_m .

Theorem 6.1 (Identity–bucket divisibility rigidity). *Suppose a finite family satisfies*

$$\sum_m a_m d_m(F) = \mathbf{1}_{F=0} \quad (|F| \leq B). \quad (77)$$

Then

$$a_m = 0 \quad (1 \leq m \leq B), \quad \sum_{m>B} a_m = 1. \quad (78)$$

Conversely, (78) is sufficient.

Proof. Evaluate (77) at each $1 \leq n \leq B$:

$$0 = \sum_{m|n} a_m. \quad (79)$$

At $n = 1$ this gives $a_1 = 0$. Increasing induction on n then gives $a_n = 0$, since every proper divisor coefficient has already vanished. At $F = 0$, all divisibility columns equal one, so the remaining coefficients sum to one. If $m > B$, then $d_m = \mathbf{1}_{F=0}$ on the window, proving the converse. $\square$

Thus no linear combination of window–internal completion indicators can recover exact equality. Every effective coefficient in an exact rule is already attached to a no–wrap modulus. This is a no–go for that particular coefficient class, not for phase measurements or for relations among the actual physical columns.

6.2 How quotient tomography implements no–wrap recovery

The projector in theorem 3.2 may be written as

$$\mathbf{1}_{M|F} \frac{1}{q} \sum_{\nu \bmod q} \exp(2\pi i \nu F / (qM)) = \mathbf{1}_{qM|F}. \quad (80)$$

Therefore it does not contradict theorem 6.1: the effective modulus qM is larger than B . What is new is its implementation. Instead of introducing qM as an additional multiplicative CRT face modulus, we retain the three–prime recombination at M and resolve its quotient by q additive phase measurements. Each phase has unit modulus, so it does not change the scalar face masses.

6.3 A finite alias–synthesis conservation law

For comparison, let

$$R = \prod_{i=1}^s p_i \quad (81)$$

be a squarefree CRT modulus with a fixed number s of odd prime factors. The full target–synthesis calculation of TPC–38 gives the exact largest squared singular value

$$\Lambda_C(R) = \prod_{i=1}^s \frac{p_i(p_i - 2)^2}{(p_i - 1)^2} = R \prod_{i=1}^s \left(1 - \frac{1}{p_i - 1}\right)^2. \quad (82)$$

The joint alias count in the window is

$$K_R = 1 + 2 \left\lfloor \frac{B}{R} \right\rfloor. \quad (83)$$

Proposition 6.2 (Alias–synthesis conservation). *If s is fixed and $\min_i p_i \rightarrow \infty$, then*

$$\boxed{\Lambda_C(R)K_R \asymp R + B.} \quad (84)$$

In particular, coefficient–blind full synthesis cannot make both the modulus cost and the joint alias count smaller than the determinant scale.

Proof. The product in (82) is bounded above and below by positive constants and tends to one. Also $RK_R \asymp R + B$. $\square$

This is a finite spectral trade, not a lower bound for the Möbius physical output. Directly increasing R reduces aliases in inverse proportion to the dominant full synthesis scale. Additive quotient tomography keeps $R = M$ in the multiplicative face calculus and pays instead for a bank of

$$q = L + 1 \asymp 1 + B/M \quad (85)$$

structured outputs. The recovery functional has unit normalized amplification, but an arithmetic theorem uniform or square–averaged over this bank is still required. Neither implementation is coefficient–free evidence of a power saving.

7 Twisted face fields and the physical recovery gate

We now return the quotient algebra to the inherited physical interface. Every statement that estimates a physical output is isolated as a conditional input.

7.1 Unimodular twists preserve the face ledgers

For $0 \leq \nu < q$, set

$$C_{\nu,N}(F) := \exp(2\pi i \nu F / (qM)) C_N(F), \quad (86)$$

$$P_{\nu,N}(F) := \exp(2\pi i \nu F / (qM)) P_N(F). \quad (87)$$

The twist depends only on the determinant. In particular,

$$|C_{\nu,N}(F)| = |C_N(F)|, \quad |P_{\nu,N}(F)| = |P_N(F)|, \quad (88)$$

$$\sum_{|F| \leq B} |C_{\nu,N}(F)|^2 = \sum_{|F| \leq B} |C_N(F)|^2, \quad \sum_{|F| \leq B} |P_{\nu,N}(F)|^2 = \sum_{|F| \leq B} |P_N(F)|^2. \quad (89)$$

The one–period identities from TPC–38 therefore imply, uniformly in unit targets and twists,

$$\sum_{|F| \leq B} |C_{\nu,N}(F)|^2 \ll K_B + 1, \quad (90)$$

$$\sum_{|F| \leq B} |P_{\nu,N}(F)|^2 \ll \frac{K_B + 1}{J}, \quad (91)$$

at the orbit–scale three–prime endpoint. Partial periods at the two ends only change the absolute constants.

These scalar identities do not transfer automatically through the coherent physical synthesis. They show that adding the phase bank does not worsen the coefficient norm in the proposed face–field estimate.

7.2 A canonical minimal–bank square–energy condition

Recall $\mathcal{Y}_\nu$ from (42). The exact analytic input tailored to the minimal tomography bank is the following.

Conditional input 7.1 (Minimal twist–bank energy). The actual regular–subpacket outputs satisfy

$$\boxed{\frac{1}{q} \sum_{\nu=0}^{q-1} \|\mathcal{Y}_\nu\|^2 \ll X^{o(1)} B_0 K_B.} \quad (92)$$

Theorem 7.2 (Conditional equality closure). *Under conditional input 7.1,*

$$\|\mathcal{R}_0\|^2 \ll X^{o(1)} B_0 K_B. \quad (93)$$

At the endpoint this is

$$\|\mathcal{R}_0\|^2 \ll X^{o(1)} Q^3 / J. \quad (94)$$

Proof. Use (45), $\mathcal{Z}_0 = \mathcal{R}_0$, and (36). $\square$

This is an exact sufficient implication, but it is not an analytic weakening of the original target. Indeed, (50) shows that the hypothesis is stronger than the desired conclusion because its left side contains $\|\mathcal{R}_0\|^2$ as a positive summand. Its value is to specify a finite structured family of completed outputs to which additive-dispersion methods could be applied, not to lower the required cancellation by algebra alone.

7.3 A face–separated sufficient input

A formulation closer to the TPC–37 prescribed–field proposal is

Conditional input 7.3 (Twist–stable full/proper fields). The outputs in (40) and (41) satisfy

$$\frac{1}{q} \sum_{\nu=0}^{q-1} \|\mathcal{Y}_\nu^C\|^2 \ll X^{o(1)} B_0 K_B, \quad (95)$$

$$\frac{1}{q} \sum_{\nu=0}^{q-1} \|\mathcal{Y}_\nu^P\|^2 \ll X^{o(1)} B_0 \frac{K_B}{J}. \quad (96)$$

Proposition 7.4 (Face stability implies the minimal gate). *Conditional input 7.3 implies conditional input 7.1.*

Proof. For each ν ,

$$\|\mathcal{Y}_\nu^C + \mathcal{Y}_\nu^P\|^2 \leq 2\|\mathcal{Y}_\nu^C\|^2 + 2\|\mathcal{Y}_\nu^P\|^2.$$

Average and use $J \rightarrow \infty$. $\square$

Uniform estimates for every ν with the corresponding right sides are stronger sufficient conditions. They would incur no extra factor q , because the final recovery uses an average. On the other hand, the two untwisted estimates proposed in TPC–37 are only the case $\nu = 0$. They do not imply either twist–stable assumption. The two–column obstruction from TPC–38, or the deletion witness in corollary 5.5, supplies abstract counterexamples.

7.4 Why this is smaller than an all-field theorem

An all-field Bessel estimate stable under arbitrary determinant restrictions would control every field $\kappa_N(F)$, hence in particular all $C_{\nu,N}, P_{\nu,N}$. The minimal twist bank asks only for the $2q$ explicit fields in (86)–(87), or more weakly for their q recombinations. It discards the L -dimensional antisymmetric kernel in (55). It is therefore a genuinely smaller operator test than the hereditary all-field theorem isolated in TPC-38.

It is not known whether the actual product-slope and projective reassembly methods are stable under this bank. Establishing precisely that stability is the new physical gate.

8 Terminal injectivity and the Möbius boundary

The terminal factorization $s = 1$ displays the arithmetic content that the twist bank must estimate. We first remove a possible combinatorial ambiguity: distinct row-alias atoms do not represent the same integer.

8.1 Exact terminal injectivity

For fixed $j \in \mathcal{J}$, define

$$n_{\alpha,j,k} := m_{\alpha}j + h + kM, \quad -L \leq k \leq L. \quad (97)$$

Theorem 8.1 (Terminal row-alias injectivity). *Assume (27) and, as at the endpoint,*

$$\max_{\alpha,\beta} |m_{\alpha} - m_{\beta}| < M. \quad (98)$$

For every fixed j , the map

$$(\alpha, k) \mapsto n_{\alpha,j,k} \quad (99)$$

is injective.

Proof. If $n_{\alpha,j,k} = n_{\beta,j,k'}$, then

$$(m_{\alpha} - m_{\beta})j = (k' - k)M. \quad (100)$$

Every prime factor q_i of M is larger than the positive integer j , so $(j, M) = 1$. Thus $M \mid m_{\alpha} - m_{\beta}$. By (98), the two row integers are equal. Source separation gives $\alpha = \beta$, and then (100) gives $k = k'$. $\square$

At the endpoint, $m_{\alpha} \asymp Q$ and $M/Q \rightarrow \infty$, so (98) is automatic after the dyadic constants are fixed. The theorem says that no hidden target multiplicity is responsible for the terminal difficulty.

8.2 The atomic diagonal is already at the target scale

With a terminal cutoff Φ_0 , put

$$b_{\alpha,j,k} := -\mu(n_{\alpha,j,k}) \log(n_{\alpha,j,k}) \Phi_0(n_{\alpha,j,k}), \quad (101)$$

interpreting the term as zero outside the positive terminal support. A left terminal slice, including the equality alias $k = 0$, has the form

$$\mathcal{T}_{\gamma,j}^L = \sum_{\alpha} \sum_{k \in \mathcal{I}_B} c_{\alpha,\gamma,j}^L b_{\alpha,j,k}, \quad (102)$$

where

$$c_{\alpha,\gamma,j}^L = \gamma_\alpha^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j). \quad (103)$$

The right slice is symmetric. The shift changes the terminal atom b , not the inherited row-time multiplier, and the deterministic multiplier ledger gives

$$\sup_{\alpha,\gamma,j} |\mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j)| \ll X^{o(1)}. \quad (104)$$

This is only a pointwise mask, residue, and smooth-weight bound; it is not the unproved moving-shift Hilbert output estimate. All cutoffs and logarithms are absorbed into $X^{o(1)}$.

For the ν -th twist-bank member, each summand additionally carries $\omega^{\nu k}$. This factor has modulus one and disappears on the atomic diagonal, so the same diagonal bound holds after averaging over ν .

Expanding the two squared energies gives

$$\|\mathcal{T}^L\|_2^2 + \|\mathcal{T}^R\|_2^2 = \mathcal{D}_{\text{term}} + \mathcal{O}_{\text{term}}, \quad (105)$$

where $\mathcal{D}_{\text{term}}$ contains only $(\alpha, k) = (\beta, k')$, and $\mathcal{O}_{\text{term}}$ contains every other pair.

Proposition 8.2 (Terminal diagonal closure). *The atomic diagonal satisfies*

$$\boxed{\mathcal{D}_{\text{term}} \ll X^{o(1)} Q^2 J K_B.} \quad (106)$$

For $B \asymp X$,

$$\mathcal{D}_{\text{term}} \ll X^{o(1)} \frac{Q^3}{J}. \quad (107)$$

Proof. There are $O(X^{o(1)}Q)$ opposite rows and $O(J)$ orbit times. For each of the K_B aliases, use $\sum_\alpha |\gamma_\alpha^{(i)}|^2 \ll X^{o(1)}Q$ and the bounded multiplier. This proves (106) for both sides. Finally, (36) gives (107). $\square$

Together with [theorem 8.1](#), this closes the terminal collision and atomic diagonal ledger. The remaining term $\mathcal{O}_{\text{term}}$ is genuine coefficient correlation:

(i) if $\alpha = \beta$ and $k \neq k'$, it contains

$$\mu(n_{\alpha,j,k})\mu(n_{\alpha,j,k'}), \quad n_{\alpha,j,k} - n_{\alpha,j,k'} = (k - k')M; \quad (108)$$

(ii) if $\alpha \neq \beta$, the physical row provenance contributes the additional factors $\mu(d_\alpha)\mu(d_\beta)$, producing the affine four-Möbius quadratic form isolated in TPC-34 and TPC-38.

8.3 What minimal folding retains

Let

$$C_{k,k'} = \langle \mathcal{Z}_k, \mathcal{Z}_{k'} \rangle. \quad (109)$$

The minimal twisted energy is exactly

$$E_{\text{fold}} = C_{0,0} + \sum_{r=1}^L (C_{r,r} + C_{r-q,r-q} + 2 \operatorname{Re} C_{r,r-q}). \quad (110)$$

The diagonal entries in this formula still contain cross-row terms at a fixed alias. In addition, each folded pair has

$$r - (r - q) = q, \quad qM \in (B, B + M], \quad (111)$$

so its cross term contains a long-gap Möbius correlation.

The twist

$$\exp(2\pi i\nu(su - N_\alpha(j))/(qM)) = \omega^{\nu k} \quad (112)$$

is constant across all rows at fixed k . It therefore does not oscillate the fixed-alias cross-row form. The two members of a folded pair have the same phase modulo q , so that correlation also survives. Most importantly, $C_{0,0} = \|\mathcal{R}_0\|^2$ remains in (110) without alteration.

8.4 Comparison with short progression variance

The closest general theorem for scalar multiplicative functions in this range is the variance estimate of Klurman et al. [3, Theorem 1.5]. After subtraction of one most-pretentious character, it gives for almost every suitable modulus $m = o(x)$ a bound of schematic size

$$\varepsilon \varphi(m) \left(\frac{x}{m} \right)^2 \quad (113)$$

over the reduced residue classes; under GRH the exceptional-modulus restriction is removed for typical moduli. Specializing only the scale to $m = M$, $x \asymp X$, and $K = X/M \asymp K_B$, and using $\varphi(M) \asymp M$, the right side has order

$$\varepsilon MK^2, \quad (114)$$

whereas the scalar atomic diagonal has order MK . Thus this black box would need $\varepsilon \ll K^{-1}X^{o(1)}$ to reach the diagonal scale. At the TPC endpoint, $K = X^{1/400+o(1)}$, while the available savings are logarithmic rather than this fixed power. The Möbius-specific adaptation discussed by those authors likewise gives a logarithmic saving, not K^{-1} .

This comparison is deliberately limited. The cited theorem averages reduced residue classes, removes a rank-one pretentious component, and is unconditional for almost all moduli. It does not contain the physical row mask, the row synthesis, or the four-Möbius form in $\mathcal{O}_{\text{term}}$. GRH changes the exceptional-modulus statement, not the missing factor K . Hence no existing scalar progression theorem is imported here as a proof of [conditional input 7.1](#).

9 Partial, redundant, and random twist banks

The minimal regular polygon is exact but has no redundant measurement. This section records what is lost under partial sampling and what random sampling can provide.

9.1 The nonzero bank restores the alias loss

Remove the zero column and define

$$\mathcal{Y}_\nu^* := \sum_{\substack{-L \leq k \leq L \\ k \neq 0}} \omega^{\nu k} \mathcal{Z}_k. \quad (115)$$

The untwisted nonzero aggregate required by the two-sum route of TPC-38 is

$$\mathcal{Y}_0^* = \sum_{k \neq 0} \mathcal{Z}_k. \quad (116)$$

Since $\nu = 0$ is one member of the bank,

$$\boxed{\|\mathcal{Y}_0^*\|^2 \leq (q-1) \left(\frac{1}{q} \sum_{\nu=0}^{q-1} \|\mathcal{Y}_\nu^*\|^2 \right)}. \quad (117)$$

Indeed, additive orthogonality gives $\sum_{\nu=0}^{q-1} \mathcal{Y}_\nu^* = 0$. Cauchy applied to the other $q - 1$ values then yields $q \|\mathcal{Y}_0^*\|^2 \leq (q - 1) \sum_{\nu} \|\mathcal{Y}_\nu^*\|^2$, which is the displayed bound. The factor $q - 1 \asymp K_B$ is sharp: the nonzero residue coefficients parametrize every zero-mean bank list. Thus applying Parseval only to the nonzero aliases and then returning to the one untwisted aggregate recreates the same endpoint order as the finite alias loss. The gain in [theorem 3.4](#) comes from averaging the *completed outputs themselves*, with the zero column included and isolated by the quotient projector.

9.2 Redundant cyclic banks

Every integer $r > L$ gives the exact recovery formula

$$Z_0 = \frac{1}{r} \sum_{\nu=0}^{r-1} Y_\nu^{(r)}, \quad (118)$$

because the zero residue class remains a singleton. The spectrum in [theorem 4.2](#) quantifies the redundancy for $L < r \leq 2L + 1$. At $r = 2L + 1$, all aliases are separated and the bank energy is the raw trace in [\(56\)](#); this is strictly stronger than necessary for equality recovery.

Using $r > L + 1$ may be helpful if an analytic argument has a small number of exceptional twists. Exact recovery from an arbitrary retained subset is still governed by the moment system [\(64\)](#); sample count alone does not control its conditioning. Clustered nodes may make the recovery weights large, while the complete regular polygon has the optimal amplification proved in [proposition 5.3](#).

9.3 Random phase averaging

Let $\theta_1, \dots, \theta_m$ be independent and uniform on $[0, 1)$, and define

$$\hat{Z}_0 := \frac{1}{m} \sum_{j=1}^m \sum_{k=-L}^L \exp(2\pi i k \theta_j) Z_k. \quad (119)$$

Proposition 9.1 (Exact random mean-square error). *For every finite Hilbert column family,*

$$\mathbb{E} \hat{Z}_0 = Z_0, \quad (120)$$

$$\mathbb{E} \|\hat{Z}_0 - Z_0\|^2 = \frac{1}{m} \sum_{\substack{-L \leq k \leq L \\ k \neq 0}} \|Z_k\|^2. \quad (121)$$

Proof. The expectation of $\exp(2\pi i k \theta)$ is $\mathbf{1}_{k=0}$, proving [\(120\)](#). Independence reduces the variance of the sample mean by m , and orthogonality in one sample removes all terms with distinct nonzero frequencies, proving [\(121\)](#). $\square$

Random sampling is always unbiased. It becomes quantitatively useful only when a nonzero-alias trace is available. Without that input, it gives no coefficient-free deterministic recovery. The structured regular polygon is exceptional: it obtains zero error with only $L + 1$ samples by exact algebra rather than probabilistic concentration.

9.4 Partial control is not enough

The deletion no-go in [corollary 5.5](#) has an analytic consequence. Bounds for all but one member of the minimal bank, no matter how strong, do not control Z_0 in the abstract column class. A proof of the physical route must therefore do one of the following:

- (i) control the complete minimal bank in square average;
- (ii) add redundancy and solve the resulting stable moment system after deleting the exceptional twists; or
- (iii) estimate each exceptional output separately.

This finite issue should be settled before any exceptional–frequency argument is used in the Möbius analysis.

10 The corrected continuation after quotient tomography

The equality–recovery problem is now algebraically closed. The surviving question is whether the physical coefficient system admits the required twist–stable energy estimate.

10.1 What is unconditional

The paper proves, without a Möbius hypothesis:

- (i) the exact quotient identity (39);
- (ii) exact recovery of $\mathcal{Z}_0$ from $L + 1$ recombined twists with unit normalized recovery amplification;
- (iii) Hilbert–valued folded Parseval and the complete tomography spectrum for every $L < r \leq 2L + 1$;
- (iv) minimality of the $L + 1$ sample count among arbitrary phase nodes and complex weights;
- (v) optimal noise amplification and rigidity of positive minimal rules;
- (vi) the deletion no–go, the random estimator identity, and the exact relation to raw trace and alias Gram gates;
- (vii) rigidity of exact recovery by pure divisibility indicators;
- (viii) the finite alias–synthesis conservation law;
- (ix) terminal row–alias injectivity and, under the inherited deterministic multiplier ledger, closure of its complete atomic diagonal at the target scale.

These results repair the logical gap in TPC–38: there is now an explicit linear functional that converts a structured family of congruence completions into exact equality.

10.2 Route T: the minimal recombined bank

The canonical minimal–bank continuation isolated here is precisely [conditional input 7.1](#):

$$\frac{1}{q} \sum_{\nu \bmod q} \|\mathcal{Z}_\nu\|^2 \ll X^{o(1)} B_0 K_B. \quad (122)$$

This tests only the $q = L + 1$ Fourier directions needed for equality. It allows arbitrary energy in the antisymmetric folded kernel (55). It is therefore the smallest exact tomographic gate constructed in this paper.

The terminal analysis shows what must be proved: the atomic diagonal is already acceptable, while fixed–alias cross–row terms, the equality column, and qM –gap folded correlations remain. A useful proof would have to exploit the actual Möbius/source coefficients, not only additive orthogonality in the alias index.

10.3 Route F: twist–stable full/proper fields

The stronger face formulation (95)–(96) is closer to the finite CRT architecture. The twist leaves every coefficient mass unchanged, so no new period factor appears on the input side. What is unknown is whether the coherent physical synthesis respects those scalar ledgers uniformly or in square average over ν .

A theorem on all admissible coefficient fields would imply Route F, but would be unnecessarily strong. The present target is only a single regular polygon orbit of the two prescribed fields.

10.4 Route S: direct superwindow completion

The divisibility rigidity theorem says that exact recovery by completion indicators requires a modulus larger than B . One may therefore enlarge the multiplicative CRT modulus itself until there are no aliases. Within the fixed–factor–count, large–prime regime of the conservation law (84), the dominant full synthesis scale then grows to the determinant scale. This is an alternative physical architecture, not a consequence of the additive route. Optimizing the factorization of such a superwindow modulus and its zero–face range is left to a subsequent layer.

10.5 Viable outcomes for the next paper

Positive twist–stable theorem.

Prove (122), or the stronger face estimates, for the actual packet. This would close the equality–recovery stage at the inherited energy scale; it would still feed into the remaining scalar prime–pair reductions rather than by itself prove a prime–pair asymptotic.

Partial arithmetic theorem.

Prove the required estimate for a terminal sublayer, a restricted family of rows, or an averaged choice of the auxiliary triple. The exact injectivity and diagonal ledger make any genuine off–diagonal saving separately visible and publishable.

Scoped twist obstruction.

Show that a declared class of additive dispersion, large–sieve, or positivity–preserving estimates cannot reach (122). Such a theorem would rule out that method class, not prove that the physical Möbius form is large.

10.6 Parity boundary

The surviving terminal form contains two– and four–Möbius correlations. This is compatible with the classical parity obstruction for sieve methods [1, 2]. Quotient tomography neither proves nor disproves a parity–breaking estimate. It proves that equality recovery itself is no longer an undefined step: the remaining statement is the explicit twist–stable physical energy theorem in (122).

11 Exact finite certificate and claim ledger

The script `experiments/tpc39_certificate.py` is a deterministic finite certificate for the algebraic identities in this paper. It uses only integers and exact rational Hilbert vectors. Roots of unity are never approximated: every phase average is evaluated by exact cyclic orthogonality. The script has no random input, floating–point literal, true–division node, external package, or Python `assert`; every check therefore remains active under `python -O`.

The certificate exhaustively checks declared finite ranges for:

- (i) isolation of the zero alias at the sharp cyclic modulus $q = L + 1$, and separation of all aliases only at $q = 2L + 1$;
- (ii) exact zero recovery and a null witness after deleting any one twist;
- (iii) Hilbert-valued folded Parseval, collision-Gram entries, ranks, traces, eigenvectors, and all multiplicities in (53);
- (iv) the finite divisibility-indicator obstruction from theorem 6.1; and
- (v) the exact variance formulas for continuous Haar phases, the collision-free $(2L + 1)$ -grid, and the folded minimal $(L + 1)$ -grid, using exact character moments and complete finite product-space enumeration.

Running

```
python experiments/tpc39_certificate.py
python -O experiments/tpc39_certificate.py
```

produces byte-identical canonical JSON records. The frozen run passes 2,305,056 exact checks. Its certificate digest is

31ed18b3158a0aa3228441d96b04d5a7d3cb8096d669dfae0d28a5339baef633.

The SHA-256 hashes of the source and canonical JSON record are, respectively,

3527cf3d4c011df887965924b3500b4b9a645eded46e59afabb51a979282f764,
2bcce40bea2e82e8a5da849aea44af79ef0c8662b8a21f8883449979206aa7d3.

The JSON also records representative rational energies and explicit Boolean claim boundaries.

The certificate is evidence for finite algebra and implementation fidelity; it is not evidence for an asymptotic Möbius estimate. In particular, its claim ledger marks the twist-stable physical bound, uniform alias-column bound, parity breach, Hardy-Littlewood asymptotic, and twin-prime theorem as false.

References

- [1] D. R. Heath-Brown. A parity problem from sieve theory. *Mathematika*, 29(1):1–6, 1982. doi: 10.1112/S0025579300012109. URL <https://doi.org/10.1112/S0025579300012109>.
- [2] Henryk Iwaniec and Emmanuel Kowalski. *Analytic Number Theory*, volume 53 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 2004. doi: 10.1090/coll/053. URL <https://doi.org/10.1090/coll/053>.
- [3] Oleksiy Klurman, Alexander P. Mangerel, and Joni Teräväinen. Multiplicative functions in short arithmetic progressions. *Proceedings of the London Mathematical Society*, 127(2):366–446, 2023. doi: 10.1112/plms.12546. URL <https://doi.org/10.1112/plms.12546>.
- [4] Liang Wang. Orbit-sliced gram energy at the Q^3 gate: Gram-copy diagonal closure, collision savings, and the terminal Möbius boundary. Preprint, repository commit 011a3edcf3f1, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/011a3edcf3f12973c92611eeb66ea9a468cd8406/papers/tpc-34-orbit-sliced-gram-energy.
- [5] Liang Wang. Orbit-scale CRT compression and uniform gram obstructions: Product-slope character spectrum, three-modulus alias reduction, and a coefficient-specific gate. Preprint, repository commit c0f2cefc8200, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/c0f2cefc8200e80543d31783bf285c035083615c/papers/tpc-35-critical-crt-coefficient-gate.
- [6] Liang Wang. Soft projective reassembly of the physical row mask: Orthogonal GCD compression, optimal ferrers separation, and coherent product-slope spectra. Preprint, repository commit ecbd29afd77b, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/ecbd29afd77b92224aa7720c11c65444407fc0e9/papers/tpc-36-soft-projective-mask-reassembly.

- [7] Liang Wang. Coordinate-face reduction at the coherent Möbius gate: Physical zero-face absorption, jacobi near-isometry, and three-modulus energy localization. Preprint, repository commit 602144a1e7d1, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/602144a1e7d11803a3651dee987e1ad51fdaba9e/papers/tpc-37-coordinate-face-reduction.
- [8] Liang Wang. Target-coupled CRT projectors and the identity bucket: Shifted jacobi grams, alias coherence, and the equality-recovery obstruction. Preprint, repository commit 6d42c2fe53dc, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/6d42c2fe53dca917446825efdcfdd899e75596c6/papers/tpc-38-target-coupled-identity-bucket.